Albert Ankomah Dodoo

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SUMMARY

I am a machine learning researcher interested in developing adaptive and trustworthy learning systems for non-stationary environments. My research interests include adaptive and continual learning, concept drift, multimodal learning, and learning under distribution shift. My research experience spans applications in cybersecurity, financial fraud detection, and medical imaging, alongside mentoring undergraduate students in machine learning research.

EDUCATION

University of Ghana Accra, Ghana
MSc. Computer Science September 2024 – September 2025
Thesis: Adaptive Memory-Based Cuckoo Search for High-Dimensional Feature Optimization in Non-Stationary Fraud Detection Systems.

University of Ghana Accra, Ghana
BSc. Computer Science August 2019 – September 2023
Thesis: Resampled Correlation-Based Feature Descriptors: A Novel Approach to Enhancing Malware Detection Capabilities.

RESEARCH EXPERIENCE

Data Intelligence and Swarm Analytics Laboratory (DISAL) Accra, Ghana
Research Assistant August 2022 - Present

- Developed an online LSTM framework incorporating EDDM and ADWIN to detect gradual and abrupt concept drift and support adaptation to evolving fraud patterns.
- Collaborated on Vision Transformer-based SARS detection from chest CT images, contributing to work subsequently published in the *Iran Journal of Computer Science*.
- Collaborated with two researchers on a Swin-LSTM IoT intrusion detection pipeline, achieving >90% precision, recall, and F1-score across benchmark datasets.
- Developed a federated learning framework for privacy-preserving IoHT intrusion detection, achieving 95.96% F1-score with a DNN while keeping raw client data decentralized.
- Conducted a systematic literature review on network security and privacy using the PRISMA framework to explore trends since 2020.
- Mentor undergraduate thesis students through weekly research meetings, providing guidance on research question formulation, methodological pipeline design, experimental planning, and scientific writing.
- Support manuscript development from undergraduate thesis projects, including restructuring research content and refining scientific writing for publication.

Space Copy Inc. Delaware, United States
STEM Research Intern June 2025 – September 2025

- Conducted 45 simulation runs across five additive-manufacturing parameters to evaluate the effects of environmental conditions on printing performance.
- Reviewed 50+ peer-reviewed studies on on-device AI, runtime optimization, and explainability to identify design considerations for autonomous edge systems.
- Assessed embedded vision and sensor fusion methods in lunar conditions, focusing on optimization trade-offs between accuracy and efficiency.

PUBLICATIONS AND PREPRINTS

- Appati, K.J., Ziamah, B., Akrofi, H.A., & **Dodoo, A.A.** SARS Detection in Chest CT-Scan Images Using Bootstrapped ViT-B16 Model. Springer, Iran Journal of Computer Science (2025). doi.org/10.1007/s42044-025-00231-1
- **Dodoo, A.A.,** Koranteng, M., Amissah, D.K., & Appati, K.J. Resampled Correlation-Based Feature Descriptors: A Novel Approach to Enhancing Malware Detection Capabilities. Preprint at *Research Square* (2023). doi.org/10.21203/rs.3.rs-4219089/v1

- Quaicoo, R., Acheampong, R., Gyamenah, P., **Dodoo, A.A.,** Soli, M.A.T., & Appati, K.J. Adapting Triple BigGAN for Image Detection Tasks: Challenges and Opportunities. Preprint at *Research Square* (2023). doi.org/10.21203/rs.3.rs-4262097/v1
- Tetteh, L.M., Sarfo, O.Y., Amissah, D.K., **Dodoo, A.A.,** Soli, M.A.T., & Appati, K.J. (2023). Enhancing Unsupervised Signature Verification Through Advanced Feature Extraction Techniques. Preprint at Research Square (2023). doi.org/10.21203/rs.3.rs-4249016/v1

CONFERENCES AND RESEARCH ENGAGEMENT

Google AI Community Center Accra, Ghana
Invited Research Speaker September 2026

- Delivered an invited talk to graduate students and faculty on online learning, contrasting continuous model updates from streaming data with conventional epoch- and batch-based training.
- Presented my master's research on adaptive learning for non-stationary fraud detection and engaged attendees in a Q&A discussion on the methodology and findings.

Deep Learning Indaba Lagos, Nigeria
Participant & Poster Presenter August 2026

- Presented master's research on adaptive feature optimization for non-stationary fraud detection systems, receiving the Best Poster Presentation Award.
- Completed a hands-on workshop on understanding the maths behind neural networks using Google Colab.

African Computer Vision Summer School (ACVSS) Accra, Ghana
Participant & Local Organizer July 2026

- Collaborated on a medical imaging hackathon developing tumor-aware MRI super-resolution with segmentation and Monte Carlo Dropout-based uncertainty estimation.
- Improved tumor-region Dice performance by 3% over standard super-resolution on unseen patients.
- Completed advanced training in 3D computer vision and generative modeling, covering camera geometry, Structure-from-Motion, VAEs, GANs, autoregressive models, and diffusion models.

TEACHING EXPERIENCE

University of Ghana Department of Computer Science Accra, Ghana
Graduate Teaching Assistant October 2025 – Present

- Support assessment through grading of mid-semester and final examinations for 120 students.
- Conduct tutorial sessions in Advanced Data Structures and Algorithms, Intelligent Systems, and Natural Language Processing for 120 students.
- Organise and deliver biweekly LaTeX workshops for over 50 undergraduate and graduate final-year students.

University of Ghana Department of Computer Science Accra, Ghana
Teaching Assistant September 2023 – October 2024

- Led tutorial sessions for 70 students in Machine Learning, Numerical Methods, and Statistical Models and Methods for Data Science.
- Demonstrated practical applications of statistical concepts such as standardization, z-score, and measures of central tendency through Python code implementations during lectures.
- Designed quizzes for 70 students and provided feedback to enhance students' understanding.

HONORS & AWARDS

- Best Poster Presentation Award, Deep Learning Indaba, Lagos, Nigeria (August 2026).
- Overall Best Graduating Graduate Student at the Department of Computer Science (September 2025).
- DataCamp Donates X DataGlobal Hub Scholarship (October 2025).
- Overall Best Research Paper Award at the University of Ghana Postgraduate Research Workshop (July 2025).

TECHNICAL SKILLS

Programming Languages: Python | SAS | R | MATLAB | SQL
ML Frameworks: PyTorch | TensorFlow/Keras | scikit-learn
Research and Development Tools: Git | LaTeX | Jupyter Notebook | Google Colab | Kaggle Notebooks